



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

PH1006-1 – SEMICONDUCTOR PHYSICS AND PHOTONICS
 (ACT, EC, EE, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

List of constants: Velocity of light, $c = 3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h = 6.63 \times 10^{-34} \text{ Js}$,

Electron mass, $m = 9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e = 1.6 \times 10^{-19} \text{ C}$,

Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J/K}$, Mass of neutron = $1.68 \times 10^{-27} \text{ Kg}$.

Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$, Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$,

Permeability of free space, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The concept of matter wave was suggested by _____
 A) Heisenberg B) de Broglie
 C) Schrodinger D) Laplace
- de Broglie wave length of a body of mass m and kinetic energy E is given by:
 A) $\lambda = \frac{\sqrt{2meV}}{h}$ B) $\lambda = \frac{h}{meV}$
 C) $\lambda = \frac{h}{\sqrt{2meV}}$ D) $\lambda = \frac{h}{2meV}$
- Which one of the following objects, moving at the same speed, has the greatest de Broglie wavelength?
 A) Neutron B) Electron
 C) Tennis ball D) Foot ball
- If uncertainty in the position of an electron is zero, the uncertainty in its momentum would be
 A) zero B) $< h/2\lambda$
 C) $> h/2\lambda$ D) Infinite
- For a particle in the ground state in an one-dimensional potential well of width L and of infinite height, the probability of finding it will be maximum at a distance of
 A) $L/2$ from the wall B) $L/4$ from the wall
 C) $3L/4$ from the wall D) $L=0$ from the wall
- Fermi level for a metal is the
 A) Highest energy level occupied by electrons at 0°C B) Average value of all available energy levels
 C) Highest energy level occupied by electrons at 0K D) Addition of energy of all available electron energy levels
- The relationship between current density J and electric field E is
 A) $J = \sigma E$ B) $J = \sigma / E$
 C) $J = \sigma/2E$ D) $J = 1/\sigma E$
- Which of the following about Fermi-Dirac distribution is false?
 A) When $E = E_F$, the probability of finding an electron with energy equal to the Fermi energy in a metal is $1/2$ at all temperatures. B) At $T = 0 \text{ K}$ all the energy level up to E_F are occupied and all the energy levels above E_F are empty.
 C) When $T > 0 \text{ K}$, some levels above E_F are partially filled while some levels below E_F are partially empty. D) When $T = 0 \text{ K}$, some levels above E_F are partially filled while some levels below E_F are empty.

9. When a pentavalent impurity is added to a pure semiconductor, it becomes
 A) An insulator B) An intrinsic semiconductor
 C) p-type semiconductor D) n-type semiconductor
10. As the doping concentration to a pure semiconductor increases, the bulk resistance of the semiconductor
 A) Remains the same B) Increases
 C) Decreases D) Becomes zero
11. In an intrinsic semiconductor, the Fermi level
 A) Lies at the middle of the forbidden energy gap B) Is near the conduction band
 C) Is near the valence band D) May be anywhere in the forbidden energy gap
12. What happens to the Hall voltage in a conductor if the magnetic field is reversed (i.e., its direction is flipped)?
 A) The Hall voltage becomes zero. B) The Hall voltage changes sign but remains the same magnitude.
 C) The Hall voltage doubles in magnitude. D) The Hall voltage remains unchanged.
13. Which of the following polarizations is very rapid?
 A) Electronic polarization B) Ionic polarization
 C) Space charge polarization D) Orientation polarization
14. A dielectric with non-polar molecules is kept in an external electric field, then
 A) The dipole moment of each molecule is zero B) The net dipole moment of the dielectric is zero
 C) The induced electric field and the net induced dipole moment of the dielectric are in the same direction D) The induced electric field and the net induced dipole moment of the dielectric are in opposite direction
15. The following material exhibits Ferroelectricity:
 A) Iron B) Platinum
 C) Hydrogen D) Rochelle salt
16. The piezoelectric effect is the polarization of a dielectric under the influence of
 A) Light B) Mechanical stress
 C) Electrical Stress D) Heat
17. Metastable states are
 A) Ground state energy levels. B) Excited state energy levels in which electrons stay for very short interval of time.
 C) Excited state energy levels in which electron can stay for unusually long time. D) Nuclear energy states.
18. Semiconductor lasers do not require two external mirrors to form an optical cavity because
 A) the laser light required is of low power B) mere driving a minimum current in the diode is enough for laser action
 C) p-section acts as one mirror and n-section acts as the other mirror D) a pair of parallel planes cleaved or polished at a particular angle in the crystal reflect the light efficiently
19. In an optical fiber, Rayleigh scattering occurs when a photon
 A) Encounters an impurity atom in its path B) Hits the cladding
 C) Encounters sharp changes in refractive index over distances smaller than its wavelength D) Encounters a microscopic bend
20. Which fiber is preferred for long distance communication?
 A) Step index single mode fiber B) Graded index multimode fiber
 C) Step index multimode fiber D) Both single mode and multi mode fiber.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss phase velocity and group velocity.	4	L*2	1	1, 2
	b) Set up the time independent Schrodinger's wave equation for a particle moving in one dimension.	8	L2	1	1, 2
	c) Calculate the momentum of an electron and the de-Broglie wavelength associated with it if its kinetic energy is 1.5 keV.	4	L3	1	1, 2

- | | | | | | | |
|----|----|---|---|----|---|------|
| 2. | a) | Explain Heisenberg's uncertainty principle. | 4 | L2 | 1 | 1, 2 |
| | b) | Solve Schrodinger's wave equation for a particle in an infinitely deep potential well of width L and show that the energy values are quantized. | 8 | L2 | 1 | 1, 2 |
| | c) | An electron has a speed of 4.8×10^5 m/s accurate to 0.012%. With what accuracy the position of electron can be located? | 4 | L3 | 1 | 1, 2 |
| 3. | a) | What are the basic assumptions of the quantum free electron theory? | 4 | L2 | 2 | 1, 2 |
| | b) | Define the terms Fermi energy and Fermi factor. Explain the Fermi factor for different temperature conditions. | 8 | L2 | 2 | 1, 2 |
| | c) | Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 300 K. | 4 | L3 | 2 | 1, 2 |

Unit – II

- | | | | | | | |
|----|----|--|---|----|---|------|
| 4. | a) | What are extrinsic semiconductors? Explain the effect of temperature on the Fermi level in an extrinsic n-type semiconductor. | 4 | L2 | 3 | 1, 2 |
| | b) | What are intrinsic semiconductors? Describe the mechanisms of carrier generation in intrinsic semiconductors. Obtain an expression for the conductivity of an intrinsic semiconductor. | 8 | L2 | 3 | 1, 2 |
| | c) | Intrinsic silicon has a carrier concentration of $1.1 \times 10^{16} \text{ m}^{-3}$. If the mobilities of electrons and holes are $0.17 \text{ m}^2\text{v}^{-1}\text{s}^{-1}$ and $0.035 \text{ m}^2\text{v}^{-1}\text{s}^{-1}$ respectively at room temperature, compute the resistivity of silicon. | 4 | L3 | 3 | 1, 2 |
| 5. | a) | State Hall Effect: Obtain an expression for the Hall coefficient and Hall voltage of an n-type semiconductor. | 6 | L2 | 3 | 1, 2 |
| | b) | Discuss the different polarization mechanisms in dielectrics and their temperature dependence. | 6 | L2 | 4 | 1, 2 |
| | c) | The Hall coefficient of a specimen of a doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3/\text{C}$. The resistivity of the specimen is $8.93 \times 10^{-3} \text{ ohm-m}$. Find the mobility and density of the charge carrier assuming single carrier concentration. | 4 | L3 | 3 | 1, 2 |
| 6. | a) | Discuss (i) Ferroelectric effect and (ii) Piezoelectric effect. | 4 | L2 | 4 | 1, 2 |
| | b) | Explain the term internal field and derive an expression for the same in case of solids. In case of gases the internal field is just equal to the applied field, why? | 8 | L2 | 4 | 1, 2 |
| | c) | An elemental solid dielectric material has polarizability $7 \times 10^{-40} \text{ Fm}^2$. Assuming the internal field to be Lorentz field, calculate the dielectric constant for the material if the material has $3 \times 10^{28} \text{ atoms/m}^3$. | 4 | L3 | 4 | 1, 2 |

Unit – III

- | | | | | | | |
|----|----|---|---|----|---|------|
| 7. | a) | Explain the terms (i) Stimulated emission, (ii) Metastable state and (iii) Population inversion. Explain the necessity of population inversion for lasing. | 4 | L2 | 5 | 1, 2 |
| | b) | Explain the construction and working of a semiconductor laser with neat sketches. | 8 | L2 | 5 | 1, 2 |
| | c) | Calculate the acceptance angle and critical angle for the core-cladding interface when the core R.I. is 1.48 and fractional index change is 2%. | 4 | L3 | 5 | 1, 2 |
| 8. | a) | With suitable diagrams explain step index multimode fiber. | 4 | L2 | 5 | 1, 2 |
| | b) | Discuss the point-to-point optical fiber communication system and mention any three advantages of an optical fiber communication system over conventional communication system. | 8 | L2 | 5 | 1, 2 |
| | c) | A He-Ne laser emits light at a wavelength of 632.8 nm and has an output power of 2.3 mW. How many photons are emitted in each minute by this laser when operating? | 4 | L3 | 5 | 1, 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
